

On common features of the flux pumping and of the RMP shielding in tokamaks

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Abstract

We draw attention to some similarities between flux pumping and RMP shielding and discuss the implications of these similarities for the modelling of these two phenomena.

I. THE SIMILARITIES

Although flux pumping (FP) and shielding of externally produced resonant magnetic perturbations (RMPs) seem to be rather different phenomena, they have few common features which suggest that driving mechanism of these phenomena is largely the same. Respectively, modelling methods of these phenomena can also be similar what suggests that our expertise in the kinetic modelling of the RMP shielding can also be used for modelling the FP effect. Namely, we should notice the following:

- Both, FP and RMP shielding are the steady state phenomena where some mechanism generates stationary currents which, in the first case, redistribute the toroidal current so that a specific, flat safety factor profile is sustained around the magnetic axis, and, in the second case, strongly minimize the value of the resonant radial magnetic field component at the rational surfaces.
- The value of the safety factor in the region where current is created is very close to some rational number. Namely, in case of FP, $q - 1$ is typically around and even below 1 %. In case of RMPs, typical width of the shielding current is of the order of ion Larmor radius which translates to the values of the similar order (below 1%) for $q - q_{\text{res}}$ within the localization region of the shielding current.
- According to Jardin et al¹, flux pumping results from an additional loop voltage which is essentially an integral of $\mathbf{B} \cdot \nabla \Phi$, i.e. the necessary condition is the presence of the parallel electrostatic field which is absent if the perturbed equipotentials are aligned with magnetic flux surfaces. In turn, according to Markl et al², a fraction of RMP shielding currents (dominant at high collisionalities) is driven by the perpendicular misalignment electric field which is tangential to the perturbed flux surfaces (this field is algebraically related to the respective parallel electric field and it vanishes if equipotentials and flux surfaces are aligned).
- According to Krebs et al³, main contributor to the FP mechanism is the dynamo effect which drives the inductive field via the $\delta \mathbf{V} \times \delta \mathbf{B}$ term. According to Becoulet et al⁴ and by Markl et al² who obtained the same result within the kinetic theory, RMP shielding currents are driven by the dynamo effect described by the $\mathbf{V}_e \times \delta \mathbf{B}$ term

where $\mathbf{V}_e$ is the perpendicular electron fluid velocity within the flux surfaces and $\delta\mathbf{B}$ is the perturbation field (only the radial component of $\delta\mathbf{B}$ contributes).

- In both cases, despite the fact that fluid theory is used far beyond its validity range (the mean free path l_c is not small compared to the magnetic field scale R but is rather the opposite, $l_c/R \gg 1$), fluid theory often provides reasonable and even quantitatively correct results.

Based on these observations, we examine the common features in more details below. We note that in the linear RMP shielding theory, magnetic field topology is not modified by the reconnection process within most of the shielding current profile (only very near the resonant surface, negligible small islands are formed). Perturbation field leads only to the corrugation of the flux surface. Similarly, in FP cases, safety factor stays above 1, and there is no reconnection either - unstable mode only corrugates the flux surfaces leaving them embedded. Thus, if we consider a single flux surface, dynamics of electrons responsible for the parallel current is basically the same in both cases.

II. REDISTRIBUTION OF THE TOROIDAL CURRENT OVER THE RADIUS WITHIN THE FLUX PUMPING MECHANISM

For simplicity, we consider a straight cylinder tokamak model for the unperturbed equilibrium and use cylindrical coordinates $(r, \vartheta, \varphi = z/R_0)$ where the last coordinate is the normalized distance along the cylinder axis and R_0 is the major radius. We consider a separate flux surface $r = \text{const}$ where a helically closed current is formed such that contra-variant current density components are

$$j^k(r, \vartheta, \varphi) = j_{\mathbf{m}}^k(r) \cos(m\vartheta + n\varphi). \quad (1)$$

We assume that flux surface is near resonant such that $|m + nq| \ll 1$, and current flows mostly parallel to the magnetic field described by the straight field lines $B^\varphi = q(r)B^\vartheta$ in our coordinates. Respectively, main, parallel part of the current satisfies $j_{\parallel}^\varphi = qj_{\parallel}^\vartheta$. Perpendicular component which closes the divergence of this current is small and flows mainly within the same flux surface so that we can write the total current density as $j^\varphi = q_{\text{res}}j^\vartheta$ where $q_{\text{res}} = -m/n$. In case of RMP in AUG, $n = 2$ and $m = \dots, -8, -7, -6, -5$ correspond to the resonances at the edge and are rather large. In case of FP, $(m, n) = (-1, 1)$.

As one can see from Eq. (1), in the absence of the corrugation of flux surfaces by the perturbation field, such that perturbed flux surfaces are the same with unperturbed flux surfaces, perturbation current has zero toroidal average and does not lead to redistribution of axisymmetric toroidal current over the radius. We assume now that perturbation field introduces the corrugation of flux surfaces which remain circular but are meandering with the toroidal angle. Namely, perturbed flux surfaces are given by $\rho = \text{const}$ where

$$\rho = r + \Delta \cos(m\vartheta + n\varphi + \alpha) \quad (2)$$

is the flux surface label. We assume small perturbation such that $\Delta \ll r$. For the description of the perturbation current, we use the coordinate system $(\rho, \vartheta, \varphi)$, i.e. we replace only the radial variable. For simplicity we assume here $\Delta = \text{const}$ and $\alpha = \text{const}$ so that the Jacobian is unchanged, $\sqrt{g_{\text{new}}} = \sqrt{g_{\text{old}}} \partial r / \partial \rho = \sqrt{g_{\text{old}}} = r$. Then, the divergence-free current density with $j^\rho = 0$ such that current flows within the perturbed flux surface is of the form

$$j^\varphi(\rho, \vartheta, \varphi) = \frac{\rho}{r} j_{\mathbf{m}}^\varphi(\rho) \cos(m\vartheta + n\varphi), \quad j^\vartheta(\rho, \vartheta, \varphi) = -\frac{n}{m} j^\varphi(\rho, \vartheta, \varphi). \quad (3)$$

Transforming back to the cylindrical coordinates (r, ϑ, φ) and expanding over small Δ up to the linear order, we obtain the toroidal current density as

$$j^\varphi \approx j_{\mathbf{m}}^\varphi(r) \cos \phi + (\cos \alpha + \cos(2\phi + \alpha)) \frac{\Delta}{2r} \frac{\partial}{\partial r} r j_{\mathbf{m}}^\varphi(r), \quad (4)$$

where we introduced the phase $\phi = m\vartheta + n\varphi$. It can be seen that this current density contains a non-zero toroidal average (average over ϕ),

$$\overline{j^\varphi}(r) \approx \frac{\Delta \cos \alpha}{2r} \frac{\partial}{\partial r} r j_{\mathbf{m}}^\varphi(r), \quad (5)$$

which corresponds to zero toroidal current,

$$I_{\text{tor}} = 2\pi \int_0^\infty dr \, r \overline{j^\varphi}(r) = \pi \Delta \cos \alpha [r j_{\mathbf{m}}^\varphi(r)]_{r=0}^{r=\infty} = 0, \quad (6)$$

i.e. helical current leads only to the re-distribution of the average toroidal current. Thus, we should study the same problem of the formation of helical current at the corrugated flux surfaces in both, RMP shielding and flux pumping cases.

III. DRIFT-KINETIC HELICAL CURRENT

As we know from the RMP shielding problem, main shielding current is produced by the electrons which, in contrast to the ions, are well described by the local approximation (at each flux surface independently). Such a local description can be valid also in the helical core except for the cases where embedding of flux surfaces is destroyed by the reconnection, where the radial transport induced by misaligned electric potential is significant and where supra-thermal electrons are produced by strong ECRH/ECCD. We describe the electrons by the stationary drift-kinetic equation of the form

$$\mathbf{v}_g \cdot \nabla f - \hat{L}_c f = 0, \quad (7)$$

where we use total energy and magnetic moment as velocity space variables and where $\hat{L}_c$ is the linearized collision operator which conserves energy (because electron-ion energy exchange is negligible) but does not conserve the momentum (which is rapidly lost to the ions within electron-ion collisions). As we usually do in the RMP shielding modelling, we can ignore the magnetic drift in $\mathbf{v}_g$ because it is of the order of inverse aspect ratio compared to the $\mathbf{E} \times \mathbf{B}$ drift $\mathbf{v}_E$ so that $\mathbf{v}_g = v_{\parallel} \mathbf{h} + \mathbf{v}_E$ where $\mathbf{h} = \mathbf{B}/B$. The ignored magnetic drift results in usual bootstrap and Pfirsch-Schlüter currents which are, respectively, small in the RMP shielding problem compared to the current driven by $\mathbf{v}_E$ (see Ref. 5 where detailed estimates are given), and they appear to be also small in the flux pumping case as our recent calculations of bootstrap current by the NEO-2 code have shown.

As previously, we stay with the straight cylinder model used above and consider two approaches - linear and nonlinear δf . For both models, we present the distribution function in the form $f = f_0 + \delta f$ where f_0 is the unperturbed Maxwellian satisfying $\hat{L}_c f_0 = 0$. Models differ by the LHS for δf .

A. Linear δf model

We present $\mathbf{v}_g = \mathbf{v}_{g0} + \delta \mathbf{v}_g$ where $\mathbf{v}_{g0}$ is the guiding center velocity in the unperturbed field and $\delta \mathbf{v}_g$ is the perturbation such that

$$\mathbf{v}_{g0} = \mathbf{h}_0 v_{\parallel} + \frac{c \mathbf{h}_0 \times \nabla \Phi_0}{B_0}, \quad \delta \mathbf{v}_g \approx \delta \mathbf{h} v_{\parallel} + \frac{c \mathbf{h}_0 \times \nabla \delta \Phi}{B_0}, \quad (8)$$

where we skipped small correction to the drift within the flux surfaces which at the end does not contribute to the equation. Parameters of the unperturbed Maxwellian f_0 (density n_e and temperature T_e) are set to be functions of r only. The linearized drift-kinetic equation is then of the form

$$\mathbf{v}_{g0} \cdot \nabla \delta f - \hat{L}_c \delta f = -(\delta \mathbf{v}_g \cdot \nabla r) \frac{\partial f_0}{\partial r}. \quad (9)$$

As in Refs. 2 and 5, we introduce the wave vector $\mathbf{k} = m \nabla \vartheta + n \nabla \varphi$ and assume the dependence of perturbed quantities (δf , δh^k and $\delta \Phi$) on angles of the form

$$\delta a(r, \vartheta, \varphi) = \text{Re} \left(a_{\mathbf{m}}(r) e^{im\vartheta + in\varphi} \right). \quad (10)$$

Introducing $k_{\parallel} = \mathbf{h}_0 \cdot \mathbf{k}$ and $k_{\perp} = \mathbf{k} \cdot \mathbf{h}_0 \times \nabla r$, we transform the kinetic equation (9) to

$$i(k_{\parallel} v_{\parallel} + k_{\perp} v_{E0}) f_{\mathbf{m}} - \hat{L}_c f_{\mathbf{m}} = -(h_{\mathbf{m}}^r v_{\parallel} + v_{E\mathbf{m}}^r) \frac{\partial f_0}{\partial r}, \quad (11)$$

where $v_{E0} = c\Phi'_0(r)/B_0$. We split the perturbation of the potential into the aligned part $\delta\Phi^{(A)}$ and misaligned part $\delta\Phi^{(MA)}$ where the aligned part satisfies

$$(\mathbf{h}_0 + \delta \mathbf{h}) \cdot \nabla (\Phi_0 + \delta\Phi^{(A)}) \approx \delta h^r \Phi'_0(r) + \mathbf{h}_0 \cdot \nabla \delta\Phi^{(A)} = 0, \quad (12)$$

which means for the Fourier amplitudes that

$$h_{\mathbf{m}}^r \Phi'_0(r) + i k_{\parallel} \Phi_{\mathbf{m}}^{(A)} = 0. \quad (13)$$

Respectively, amplitude of the drift velocity perturbation is also split in two parts $v_{E\mathbf{m}}^r = v_{E\mathbf{m}}^{r(A)} + v_{E\mathbf{m}}^{r(MA)}$. Splitting also the solution as $f_{\mathbf{m}} = f_{\mathbf{m}}^{(A)} + f_{\mathbf{m}}^{(MA)}$, we obtain equations for the separate parts,

$$i(k_{\parallel} v_{\parallel} + k_{\perp} v_{E0}) f_{\mathbf{m}}^{(A)} - \hat{L}_c f_{\mathbf{m}}^{(A)} = -(h_{\mathbf{m}}^r v_{\parallel} + v_{E\mathbf{m}}^{r(A)}) \frac{\partial f_0}{\partial r}, \quad (14)$$

$$i(k_{\parallel} v_{\parallel} + k_{\perp} v_{E0}) f_{\mathbf{m}}^{(MA)} - \hat{L}_c f_{\mathbf{m}}^{(MA)} = -v_{E\mathbf{m}}^{r(MA)} \frac{\partial f_0}{\partial r}. \quad (15)$$

Due to

$$h_{\mathbf{m}}^r v_{\parallel} + v_{E\mathbf{m}}^{r(A)} = \frac{h_{\mathbf{m}}^r}{k_{\parallel}} (k_{\parallel} v_{\parallel} + k_{\perp} v_{E0}),$$

we can solve Eq. (14) as follows,

$$f_{\mathbf{m}}^{(A)} = -\frac{i h_{\mathbf{m}}^r}{k_{\parallel}} \frac{\partial f_0}{\partial r}, \quad (16)$$

where one should remember that $\hat{L}_c \partial f_0 / \partial r = 0$ for the energy preserving collision operator. Obviously, the aligned solution (16) is even in $v_{\parallel}$ and produces no parallel current. Thus, parallel current is driven only by the misaligned potential, as discussed in the introduction.

More explicit form of Eq. (15) which is of interest for the structure of the parallel current density is

$$i(k_{\parallel} v_{\parallel} + k_{\perp} v_{E0}) f_{\mathbf{m}}^{(\text{MA})} - \hat{L}_c f_{\mathbf{m}}^{(\text{MA})} = \frac{c f_0 E_{\perp \mathbf{m}}^{(\text{MA})}}{B_0} \left(A_1 + \frac{m v^2}{2T} A_2 \right), \quad (17)$$

where $E_{\perp \mathbf{m}}^{(\text{MA})} = -i k_{\perp} \Phi_{\mathbf{m}}^{(\text{MA})}$ is the perpendicular misalignment field, and the thermodynamic forces are

$$A_1 = \frac{1}{n_e} \frac{\partial n_e}{\partial r} + \frac{e_e}{T_e} \frac{\partial \Phi_0}{\partial r} - \frac{3}{2T_e} \frac{\partial T_e}{\partial r}, \quad A_2 = \frac{1}{T_e} \frac{\partial T_e}{\partial r}. \quad (18)$$

This form shows that for driving the electron current one needs not only the perpendicular misalignment field² (equivalently, the misaligned potential) but also the radial gradients of the unperturbed electron density and temperature and of the electrostatic potential. In case of RMPs, all these three gradients contribute the comparable amount. In case of FP with central ECRH/ECCD, electron temperature gradient is the main contributor.

B. Nonlinear δf

The above solution is sufficient for the RMP shielding problem where helical current is the only one of interest but is not sufficient for the FP problem where we need to find helical current on the perturbed flux surfaces. The ansatz now is to retain $\delta \mathbf{h}$ and $\delta \mathbf{v}_E$ also in the LHS of the drift kinetic equation and to replace the radial variable with flux surface aligned ρ (flux surface label). Problem will remain slightly non-local, but in the regimes where the misaligned potential contribution can be ignored in the LHS, problem becomes local and very similar to Eq. (15). Thus, we introduce the label ρ for the corrugated flux surfaces as follows

$$\rho = r + \text{Re} \left(\rho_{\mathbf{m}} e^{im\vartheta + in\varphi} \right), \quad \rho_{\mathbf{m}} = -\frac{i h_{\mathbf{m}}^r}{k_{\parallel}}. \quad (19)$$

This label satisfies the linearized flux surface label definition (i.e. we still assume the perturbations to be small),

$$h^{\rho} \equiv \mathbf{h} \cdot \nabla \rho \approx \delta h^r + \mathbf{h}_0 \cdot \nabla (\rho - r) = 0 \quad \Rightarrow \quad h_{\mathbf{m}}^r + i k_{\parallel} \rho_{\mathbf{m}} = 0. \quad (20)$$

It can be checked than that Eqs (13) and (16) mean that within the linear order

$$\Phi_0(r) + \delta\Phi^{(A)}(r, \vartheta, \varphi) \approx \Phi_0(\rho), \quad f_0(r, v) + \delta f^{(A)}(r, \vartheta, \varphi, v) \approx f_0(\rho, v). \quad (21)$$

Then, the nonlinear δf equation

$$\mathbf{v}_g \cdot \nabla \delta f - \hat{L}_c \delta f = -\mathbf{v}_g \cdot \nabla f_0, \quad (22)$$

which is actually equivalent to the full- f equation can be treated in the same way as in section II, i.e. replacing the radial coordinate r with ρ but keeping the same angles in the flux coordinate system $(\rho, \vartheta, \varphi)$ associated with the corrugated flux surfaces. Since by definition (20) there is no parallel particle streaming across the perturbed flux surfaces, radial guiding center velocity v_g^ρ is solely due to the $\mathbf{E} \times \mathbf{B}$ drift driven by the misaligned potential $\delta\Phi^{(MA)}$,

$$\mathbf{v}_g \cdot \nabla \delta f - \hat{L}_c \delta f = -v_E^{\rho(MA)} \frac{\partial f_0}{\partial \rho}. \quad (23)$$

In case this radial velocity can be ignored in the LHS of Eq. (23) (conditions to ignore it are discussed in the next section), problem becomes local, and we can use again Fourier expansions (10) with the replaced radial variable $r \rightarrow \rho$ and obtain for the Fourier amplitude (being now the function of ρ instead of r) formally the same equation as Eq. (15),

$$i(k_\parallel v_\parallel + k_\perp v_{E0}) f_{\mathbf{m}} - \hat{L}_c f_{\mathbf{m}} = -v_{E\mathbf{m}}^{\rho(MA)} \frac{\partial f_0}{\partial \rho}, \quad (24)$$

where all functions of r are replaced with the same functions of ρ and we have omitted the superscript (MA) on the Fourier amplitude of the distribution function because the aligned part has been anyway absorbed into f_0 . It should be noted that on the way from the exact Eq. (23) to the approximate Eq. (24) we have ignored not only the radial drift in the LHS but also a small angular variation of metrics, $k_\parallel$ and $k_\perp$, so that Fourier modes are decoupled in Eq. (24). This is already a minor problem (we ignore terms of the order of Δ/R) compared to ignoring the toroidal effects of the realistic geometry (terms of the order of r/R and even $(r/R)^{1/2}$, see the next section).

IV. VALIDITY OF FLUID AND LOCAL KINETIC MODEL, POSSIBLE INITIAL BENCHMARKING

It should be mentioned that nonlocal Eq. (23) (which actually is valid in realistic geometry) can be treated as is by the nonlinear δf Monte Carlo method. However, this is a

rather computer intensive method and it will be an overkill in case we can use a local ansatz (i.e. ignore v_g^p in the LHS). In the latter case, we can use NEO-2 code⁶ with some modifications (drive by the misaligned potential should be implemented in its quasilinear version version^{7,8} NEO-2-QL) if the corrugation of the flux surfaces is small. Such a model will naturally account the realistic geometry as well as magnetic drifts. At the same time, we should note that flux pumping is not an essentially toroidal effect, i.e. it can be modelled also in the simplified straight cylinder geometry discussed above. Effects of toroidal geometry usually do not introduce principal novelty for thermal electrons and can be ignored even at the plasma edge with RMPs (see Ref. 5). In the core, this approximation is even better because the amount of trapped particle is much smaller. Thus, as a very fast tool, one can also use the analytical treatment of Eq. (24) in terms of susceptibility functions of the code KILCA^{9,10}, as it is done in Refs 2 and 5.

Finally, let us discuss now why fluid models describe flux pumping rather well despite being formally invalid and what are the real limitations of the fluid approach. Actually the answer is above: flux pumping is essentially a cylindrical geometry effect where the realistic toroidal geometry does not introduce significant novelties (we speak about the **electron** dynamics here, but one should be cautious with **ions**). If toroidal effects are ignored, problem acquires helical symmetry, and the validity conditions of the fluid approximations become rather different from those in the neoclassical theory ($\nu^* \gg 1$ etc). There are actually two such conditions. The first one, $\omega_E \ll \nu_e$, where $\omega_E = k_\perp v_{E0}$ and ν_e is electron pitch-angle scattering frequency, is more important in the RMP shielding problem because of relatively large electric fields at the edge. If this condition is violated, Landau damping effects determine the response currents^{2,10}. Probably this condition is less restrictive in case of FP because of the smaller ω_E due to low poloidal mode number $m = -1$ and weaker electric field (though, ν_e is also smaller due to higher temperatures). The second condition is $l_c \ll R/|\Delta q|$ where $\Delta q = q - q_{\text{res}}$, i.e. the mean free path l_c should be much shorter than a modified (much longer) connection length $R/\Delta q$. If this condition is violated, field line with the length l_c can effectively cover the whole perturbed flux surface, and this will essentially deactivate radial electron transport due to Liouville's theorem (a well known effect in stellarators where despite essentially 3D geometry drift of passing particles does not lead to radial transport in the long mean free path regime). Thus, convective electron heat transport discussed in Ref. 3 will mostly disappear.

These two conditions seem to be often hold in AUG, and it makes sense to benchmark first fluid model of JOREK against the toroidal current density computed by (modified) NEO-2-QL and by the simplified analytical model based on KILCA susceptibility functions. In this benchmarking, the distribution of electrostatic potential as well as of electron density and temperature should be taken from JOREK. In this framework, also the δf Monte Carlo method should be checked (actually it should be the most accurate because there are no limitations on its validity).

For the ions, toroidal effects are more significant because of large deviations of these orbits from the flux surfaces. For the passing ions whose radial orbit width $\delta r_p \sim 2q\rho_L \sim 1.4$ cm is not too large (we take deuterium ions with energy 5 keV, magnetic field 2 T, such that Larmor radius is $\rho_L \approx 0.7$ cm), this deviation can be ignored since $\delta r_p \ll r$, where we take $r = 10$ cm to be the typical size of the helical core. However, for the trapped ions this deviation is 4 times larger, $\delta r_t \sim \delta r_p (R/r)^{1/2} \sim 6$ cm where we have used $R = 170$ cm, and this is already not a negligible quantity compared to the size of the helical core. The fraction of these particles is $(r/R)^{1/2} \sim 0.25$ is small, but we cannot immediately ignore it as we do with trapped electrons whose orbit widths are negligible, because these ions introduce a nonlocal coupling of flux surfaces. Thus, computation of the electrostatic potential Φ by JOREK should be verified by the kinetic model where charge neutrality iterations within the full- f method (method implemented in Gorilla for the stellarator geometry) will be used for the computation of Φ .

In various benchmarking with fluid model, one should probably start with reduced temperatures (to tens eV or so) in order to ensure the strict validity of the fluid model.

V. CONCLUSION

As we see, we do not always need such a universal but computer intensive tool as the δf Monte Carlo method in order to describe parallel current of the electron component using the kinetic approximation. We can still use NEO-2 and even analytical KILCA model (susceptibility functions) in many important cases. On the other hand, full- f Monte Carlo method is unavoidable in order to determine the electrostatic potential which at the end is a driving reason for the parallel current. In this part, fluid models are formally out of range, and kinetic treatment is necessary in order to obtain quantitatively accurate results. On the

other hand, kinetic corrections may be not too big.

Appendix A: Aligned potential in real geometry

Let us compute the potential perturbation aligned with the given radial perturbation field of the form

$$\delta B^r(r, \vartheta, \varphi) = \text{Re} \left(B_n^r(r, \vartheta) e^{in\varphi} \right). \quad (\text{A1})$$

We do this from the linearized alignment condition,

$$\mathbf{B} \cdot \nabla \Phi = (\mathbf{B}_0 + \delta \mathbf{B}) \cdot \nabla (\Phi_0 + \delta \Phi) \approx \mathbf{B}_0 \cdot \nabla \delta \Phi + \delta \mathbf{B} \cdot \nabla \Phi_0 = \mathbf{B}_0 \cdot \nabla \delta \Phi + \delta B^r \Phi'_0 = 0, \quad (\text{A2})$$

where $\Phi'_0 = \Phi'_0(r)$ denotes a derivative. Let us check now that there is no current for the aligned potential, i.e. that perturbation of the distribution function of the form

$$\delta f = \frac{\delta \Phi}{\Phi'_0} \frac{\partial f_0}{\partial r} \quad (\text{A3})$$

solves the linearized kinetic equation

$$\left(v_{\parallel} \frac{\mathbf{B}_0}{B_0} + c \Phi'_0 \frac{\mathbf{B}_0 \times \nabla r}{B_0^2} \right) \cdot \nabla \delta f - \hat{L}_c \delta f = - \left(v_{\parallel} \frac{\delta B^r}{B_0} + c \frac{\mathbf{B}_0 \times \nabla \delta \Phi \cdot \nabla r}{B_0^2} \right) \frac{\partial f_0}{\partial r}. \quad (\text{A4})$$

We have already checked previously that $\hat{L}_c \delta f = 0$ for the function (A3) in case of energy conserving $\hat{L}_c$. Since guiding center velocity in the l.h.s. of (A4) has no radial component, we can set there

$$\nabla \delta f \rightarrow \frac{\partial f_0}{\partial r} \frac{\nabla \delta \Phi}{\Phi'_0}$$

and obtain using (A2) the left hand side of (A4) in the form

$$\begin{aligned} \left(v_{\parallel} \frac{\mathbf{B}_0}{B_0} + c \Phi'_0 \frac{\mathbf{B}_0 \times \nabla r}{B_0^2} \right) \cdot \nabla \delta f &= \frac{1}{\Phi'_0} \frac{\partial f_0}{\partial r} \left(v_{\parallel} \frac{\mathbf{B}_0 \cdot \nabla \delta \Phi}{B_0} + c \Phi'_0 \frac{\mathbf{B}_0 \times \nabla r \cdot \nabla \delta \Phi}{B_0^2} \right) \\ &= \frac{1}{\Phi'_0} \frac{\partial f_0}{\partial r} \left(-v_{\parallel} \frac{\delta B^r \Phi'_0}{B_0} + c \Phi'_0 \frac{\mathbf{B}_0 \times \nabla r \cdot \nabla \delta \Phi}{B_0^2} \right), \end{aligned} \quad (\text{A5})$$

which obviously is the r.h.s. of (A4).

Presenting now the perturbed potential in the form

$$\delta \Phi(r, \vartheta, \varphi) = \text{Re} \left(\Phi_n(r, \vartheta) e^{in\varphi} \right), \quad (\text{A6})$$

the magnetic differential equation (A2) for $\delta \Phi$ turns into ODE for the amplitude Φ_n ,

$$B_0^\varphi \left(\iota \frac{\partial \Phi_n}{\partial \vartheta} + in \Phi_n \right) + B_n^r \Phi'_0 = 0. \quad (\text{A7})$$

We can solve this ODE by Fourier expansion over ϑ

$$\Phi_n(r, \vartheta) = \sum_m \Phi_{\mathbf{m}}(r) e^{im\vartheta}, \quad (\text{A8})$$

where $\mathbf{m} = (m, n)$ as follows

$$\Phi_{\mathbf{m}} = \frac{i\Phi'_0}{\iota m + n} \left(\frac{\delta B^r}{B_0^\varphi} \right)_{\mathbf{m}}, \quad \left(\frac{\delta B^r}{B_0^\varphi} \right)_{\mathbf{m}} = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\vartheta e^{-im\vartheta} \frac{B_n^r(r, \vartheta)}{B_0^\varphi(r, \vartheta)}. \quad (\text{A9})$$

A useful formula,

$$\frac{\delta B^r}{B_0^\varphi} = \frac{1}{\psi'_{\text{tor}}(r)} \left(\frac{\partial \delta A_\vartheta}{\partial \varphi} - \frac{\partial \delta A_\varphi}{\partial \vartheta} \right) = \frac{\iota(r)}{\psi'_{\text{pol}}(r)} \left(\frac{\partial \delta A_\vartheta}{\partial \varphi} - \frac{\partial \delta A_\varphi}{\partial \vartheta} \right), \quad (\text{A10})$$

where $\psi_{\text{tor}} = A_{0\vartheta}$ is the normalized toroidal flux and $\psi_{\text{pol}} = -A_{0\varphi}$ is the normalized poloidal flux, shows that Fourier coefficients of the ratio B_n^r/B_0^φ in (A9) up to the flux functions are the Fourier coefficients of the vector potential perturbations. Each harmonic $\mathbf{m}$ of such an expansion is resonant only at its own flux surface where $\iota m + n = 0$ and nowhere else. Thus, it is convenient to take in the examples for the toroidal case with aligned potential radial perturbations in the form of a single harmonic of vector potential leading to

$$\delta B^r = B_0^\varphi \text{Re} \left(e^{im\vartheta + in\varphi} \left(\frac{\delta B^r}{B_0^\varphi} \right)_{\mathbf{m}} \right), \quad \delta \Phi = \text{Re} \left(e^{im\vartheta + in\varphi} \frac{i\Phi'_0}{\iota m + n} \left(\frac{\delta B^r}{B_0^\varphi} \right)_{\mathbf{m}} \right). \quad (\text{A11})$$

Using

$$\sqrt{g} = \frac{\psi'_{\text{tor}}}{B_0^\varphi}, \quad (\text{A12})$$

we obtain

$$\begin{aligned} \mathbf{B}_0 \times \nabla \delta \Phi \cdot \nabla r &= (\mathbf{B}_0 \times \nabla \delta \Phi)^r = \frac{1}{\sqrt{g}} \left(B_{0\vartheta} \frac{\partial \delta \Phi}{\partial \varphi} - B_{0\varphi} \frac{\partial \delta \Phi}{\partial \vartheta} \right) \\ &= \frac{B_0^\varphi}{\psi'_{\text{tor}}} \text{Re} (i\Phi_{\mathbf{m}} (nB_{0\vartheta} - mB_{0\varphi}) e^{im\vartheta + in\varphi}) \\ &= \frac{\Phi'_0}{\psi'_{\text{tor}}} \frac{mB_{0\varphi} - nB_{0\vartheta}}{\iota m + n} B_0^\varphi \text{Re} \left(\left(\frac{\delta B^r}{B_0^\varphi} \right)_{\mathbf{m}} e^{im\vartheta + in\varphi} \right) \\ &= \frac{\Phi'_0}{\psi'_{\text{tor}}} \frac{mB_{0\varphi} - nB_{0\vartheta}}{\iota m + n} \delta B^r. \end{aligned} \quad (\text{A13})$$

Thus, the right hand side (source term) of the kinetic equation (A4) takes the form

$$\begin{aligned} - \left(v_{\parallel} \frac{\delta B^r}{B_0} + c \frac{\mathbf{B}_0 \times \nabla \delta \Phi \cdot \nabla r}{B_0^2} \right) \frac{\partial f_0}{\partial r} &= - \left(v_{\parallel} + \frac{c}{B_0} \frac{\Phi'_0}{\psi'_{\text{tor}}} \frac{mB_{0\varphi} - nB_{0\vartheta}}{\iota m + n} \right) \frac{\delta B^r}{B_0} \frac{\partial f_0}{\partial r} \\ &= - \left(v_{\parallel} + \frac{\iota c}{B_0} \frac{\Phi'_0}{\psi'_{\text{pol}}} \frac{mB_{0\varphi} - nB_{0\vartheta}}{\iota m + n} \right) \frac{\delta B^r}{B_0} \frac{\partial f_0}{\partial r}. \end{aligned} \quad (\text{A14})$$

This r.h.s. should result in zero response current in the absence of magnetic drifts.

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